ATSG: Adaptive Token Linking With Segment Anything Model Guidance for Weakly Supervised Remote Sensing Image Semantic Segmentation

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***Abstract*—Semantic segmentation of remote sensing images is vital for applications such as urban planning and disaster monitoring. However, the high cost of pixel-level annotations often results in limited labeled data, necessitating weakly supervised learning approaches. Unlike natural images, remote sensing images typically contain numerous small objects with substantial intraclass variations and high interclass similarities, which pose challenges for generating high-quality pseudolabel. In addition, while vision transformers (ViTs) have been integrated into weakly supervised semantic segmentation (WSSS) for their global modeling capabilities, they are prone to oversmoothing in dense scenes, which impedes model learning. To address these challenges, we propose two novel modules: the adaptive token linking (Adalink) module and the segment anything model (SAM)-guided boundary refiner (SGBR) module. First, Adalink employs a dynamic aggregation mechanism to analyze semantic diversity across the middle layer of ViT, adaptively selecting the feature layer and constructing token correlation graphs. It leverages a self-supervised encoder to extract hierarchical token relationships and to supervise pseudolabel generation, thereby reducing erroneous activations in cluttered scenes and improving pseudolabel quality. Second, SGBR utilizes the zero-shot segmentation capability of the SAM to refine segmentation results by incorporating object and boundary priors from the output images, significantly enhancing the completeness of small object segmentation and overall accuracy. Extensive experiments on the ISPRS Potsdam, ISPRS Vaihingen, and iSAID datasets demonstrate that our method achieves state-of-the-art (SOTA) performance and exhibits strong practical value in processing complex remote sensing scenes. Code will be available at https://github.com/zhangyifan25/ATSG**

***Index Terms*—Remote sensing images, segment anything model (SAM), weakly supervised semantic segmentation (WSSS).**

I. Introduction

SEMANTIC segmentation [1] is a fundamental task in computer vision that assigns a semantic class label to each pixel in an image, enabling fine-grained understanding and interpretation. Remote sensing image segmentation, in particular, has broad applications in urban planning, natural disaster monitoring, military reconnaissance, and related fields [2], [3].

Advances in remote sensing technology have made high-resolution image acquisition more accessible, and there are various modal datasets available, such as hyperspectral data, where there are many excellent deep learning models. SimPoolFormer [4] uses transformer-based pooling for spectral spatial feature extraction. EfficientNet [5] is known for its scalable and efficient architecture, while Tri-CNN [6] uses a three-branch structure for multiscale representation, setting a new benchmark in processing high-dimensional spectral data. However, the high resolution and complex scenes of such images, often containing numerous small objects and dense instances, make pixel-level annotation extremely laborious, requiring specialized expertise in geography or surveying. To mitigate reliance on expensive pixelwise labels, researchers have explored weakly supervised learning [7], [8], using alternative annotations such as image-level labels [9], [10], scribbles [11], [12], and bounding boxes [13], [14]. Among these, image-level labels, which only indicate the presence or absence of object classes without spatial information, are the most cost-effective but also the most challenging to utilize effectively. In this work, we focus on weakly supervised semantic segmentation (WSSS) with image-level labels, addressing its unique difficulties in remote sensing contexts.

Although WSSS has achieved remarkable progress in natural images—with techniques such as class activation map (CAM) [15], [16], self-supervision [17], and adversarial networks [18] that significantly reduce the reliance on pixel-level annotations and approach fully supervised performance on benchmarks such as PASCAL VOC [19] and COCO [20]—applying WSSS to remote sensing imagery remains challenging. Unlike natural images, remote sensing data exhibit more complex scenes, wider variations in object scales, and higher densities of small objects. These characteristics introduce noise in pseudolabel generation [21] and degrade segmentation accuracy, leaving existing weakly supervised methods insufficient for practical remote sensing applications. Current WSSS approaches generally follow two training paradigms: single-stage and multistage [22]. Multistage methods decompose the task into separate phases: first generating pixel-level pseudolabel with an auxiliary model

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and then training a segmentation network using these pseudolabel [9], [23]. Although this approach often achieves higher accuracy through iterative pseudolabel refinement, it suffers from a lower training efficiency [24], [25]. In contrast, single-stage methods train an end-to-end model directly on weakly annotated data, eliminating intermediate steps and improving computational efficiency, though at the cost of increased optimization difficulty. In this work, we propose a single-stage vision transformer (ViT)-based [26], [27] model for remote sensing WSSS. To mitigate ViT's oversmoothing problem [28], we supervise final-layer tokens using a middle layer. Unlike previous approaches that rely on fixed middle layer [28], [29], [30], we introduce a dynamic aggregation mechanism that evaluates token-link confidence across layers to adaptively select the most informative features. In addition, we employ a self-supervised encoder to extract tokenwise topological relationships, enhancing pseudolabel supervision. Furthermore, recognizing that remote sensing images are captured from an overhead perspective with distinct object boundaries, we integrate the segment anything model (SAM) [31], a foundational and promptable model trained on a massive dataset for generalizable image segmentation, to automatically extract boundary and object priors without manual annotation. SAM is currently widely promoted. Rezvan et al. [32] utilized SAM to achieve precise, automated segmentation of rice seedlings, while Trotter et al. [33] employed SAM to address the challenge of instance segmentation for marine benthic organisms, thereby advancing automated biodiversity monitoring. SAM has demonstrated powerful performance across these diverse applications. SAM's zero-shot segmentation capability serves as a "shape lock," improving object completeness and small-target localization to significantly boost segmentation accuracy. By unifying these components in a single-stage framework, our method establishes an efficient and robust WSSS pipeline. Our key contributions are given as follows.

1) To address the subtle interclass differences and significant intraclass variations in remote sensing images, we propose the adaptive token linking (Adalink) module. It incorporates a dynamic aggregation mechanism to adaptively select the middle layer, combined with a self-supervised encoder to compute tokenwise affinity. This effectively generates a high-quality pseudolabel.
2) To tackle the challenges of locating small objects and handling fragmented regions in remote sensing imagery, we propose the SAM-guided boundary refine (SGBR). By leveraging the SAM to extract object and boundary priors, it refines the segmentation output with structural constraints, thereby enhancing object integrity and spatial coherence.
3) With our optimized framework, the proposed method achieves state-of-the-art (SOTA) performance on the ISPRS Potsdam, ISPRS Vaihingen, and iSAID datasets.

II. Related Work

A. Single Stage WSSS

Single-stage training has gained increasing attention for its computational efficiency. In typical single-stage WSSS approaches [34], [35], image-level labels are first utilized for image classification [36], followed by object region localization through CAM to generate pseudolabel [37]. These pseudolabel then supervise the segmentation head for the final prediction. However, a critical limitation persists: CAM tends to activate only the most discriminative object regions, resulting in an incomplete pseudolabel that degrades segmentation performance. This has driven recent research to focus on improving pseudolabel quality. Several innovative approaches have emerged to address this challenge. DuPL proposes a progressive dual-student framework [38], where parallel networks mutually supervise each other using multilevel features while adaptively integrating high-confidence pseudolabel. Toco introduces token contrastive learning with ViT [28], exploiting a semantically rich middle layer to expand CAM coverage through global context, thereby generating more comprehensive pseudolabel. WeCLIP+ [39] employs contrastive vision-text alignment and dynamic prompt learning to enhance text encoding, improving region localization through cross-modal correspondence [40], [41] while reducing annotation dependency. AFA [42] refines initial seed regions by adaptively fusing multilevel feature affinities. While these methods have significantly advanced WSSS for natural images, they demonstrate limited generalization capability for remote sensing data due to its distinctive characteristics of complex scenes, scale variation, and dense small objects.

B. WSSS for Remote Sensing Image

Remote sensing images and natural images exhibit significant differences, leading to poor performance of WSSS models trained on natural images when applied to remote sensing data. The low intraclass similarity, such as different roof colors of buildings due to varying materials and lighting conditions, and high interclass similarity, such as nearly identical spectral characteristics between gray concrete building roofs and concrete roads or minimal color and texture differences between low green vegetation and trees, introduce noise during the pseudolabel generation process. Recent approaches have attempted to address these issues: OME [43] focuses on data preprocessing by optimizing class labels to filter out unreliable training samples. Lu et al. [44] employ self-supervised learning to enhance feature representation and propose an improved loss function incorporating gradient clipping and label smoothing to mitigate label noise. CTFA [29] introduces token-level and cls-token contrastive learning [45] to encourage more complete target region activation in CAM. While these methods have advanced WSSS for remote sensing imagery, conventional CNNs [46], [47] and fixed-layer ViT architectures remain limited in handling the inherent intraclass diversity and interclass similarity of remote sensing targets, leading to unstable pseudolabel confidence. Additional challenges arise from blurred boundaries and missed small targets in overhead imagery. Existing solutions present notable limitations: CRF [48], while effective for modeling spatial context and refining boundaries, suffers from high computational complexity and dependence on manually-designed constraints, restricting its generalization capability.

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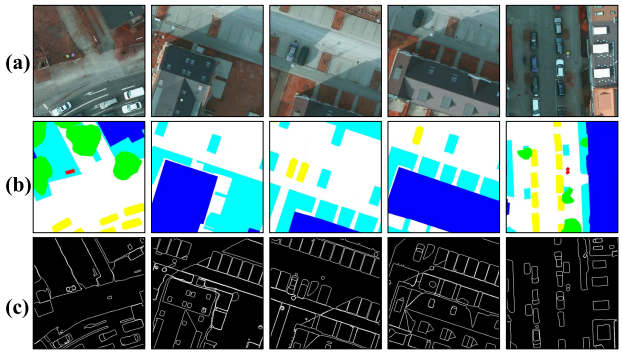


Fig. 1. Oversegmentation of SAM. A car or a building that is divided into multiple pieces. (a) Input image. (b) Ground truth. (c) Boundary prior.

C. Segment Anything Model

The emergence of large models has facilitated their increasing integration into various algorithms for performance enhancement. The SAM, as a general-purpose large-scale segmentation model, demonstrates remarkable zero-shot transfer capability [49], enabling segmentation through interactive prompts such as points or bounding boxes. In this work, we utilize its fully automatic grid-based segmentation mode without requiring manual intervention. Notably, SAMRS [50] employed SAM to construct a large-scale remote sensing segmentation dataset and successfully processed over 100 000 images. The resulting dataset provides comprehensive object classes, locations, and instance information [51], offering an innovative solution to the annotation challenges in remote sensing. Ma et al. [52] developed an SAM-based semantic segmentation framework for remote sensing images, introducing novel object target and boundary concepts. Through boundary-preserving loss and target consistency loss, their approach significantly improved segmentation performance. S2C [53] integrated prototype learning with SAM for natural images, aligning class features with their respective prototypes while increasing interclass separation, thereby enhancing CAM quality in WSSS.

Although these methods have demonstrated success in dataset generation, remote sensing semantic segmentation, and weakly supervised segmentation of natural images, to our knowledge, no existing work has yet applied SAM to WSSS of remote sensing imagery. Furthermore, SAM exhibits an inherent tendency for oversegmentation [54], often dividing single targets into multiple fragmented regions rather than producing complete semantic objects. As shown in Fig. 1, this phenomenon is evident in the boundary prior information generated by SAM, where vehicles are segmented into discontinuous parts. To address this challenge, we introduce a solution that integrates confidence scoring with dual-threshold processing.

III. Method

A. Overview

The overall architecture of our proposed network is illustrated in Fig. 2. In the classification network, we only need image-level labels and use CAM to obtain the location information of the target to generate pseudolabel, which are used to guide the segmentation head to segment. The whole process also includes the technology of optimizing the quality of the pseudolabel. The specific techniques are described in Section III-B and III-C. First, input the image $I \in \mathbb{R}^{H\times W\times 3}$, where H and W represent the height and width of the image. Our framework employs a ViT-B-based encoder–decoder architecture for segmentation, uses the ViT encoder to divide the image into $N * N$ patches, a total of L, where $L = (H \times W)/N^2$, then adds position encoding to get the token sequence, and then sends it to the transform encoder for processing. The feature map $F \in \mathbb{R}^{hw\times d}$ is obtained after image feature extraction. After being extracted by the classification encoder, its CAM is calculated from the weighted sum of the feature map and the classification layer weight $W_c \in \mathbb{R}^{c\times d}$, and c is the number of classes. Specifically, for class c, CAM is reconstructed into a 2-D spatial form through a linear combination of the feature map F and the corresponding class weight W_c, followed by ReLU activation and normalization. Therefore, CAM for class c can be expressed as follows:

$$\mathrm{CAM}_c = \frac{\mathrm{ReLU}\left(M_c\right)}{\mathrm{Max}\left(\mathrm{ReLU}\left(M_c\right)\right)}, \quad M_c = \sum_i W_i^c F_i. \tag{1}$$

After obtaining the cam, we can convert it into a pseudolabel P through the threshold value. We set a low threshold value τ_1 and a high threshold value τ_2 to distinguish the foreground background (BG) and uncertain areas, where $0 < \tau_1 < \tau_2 < 1$. For each element at the position (x, y), the element below the low threshold value will be regarded as the BG and marked as 0. For the element above the high threshold value, we will take the class with the largest activation value. The value in the middle of the threshold is the uncertain region, and we label it with a pseudolabel of 255. In subsequent model training, we ignore the uncertain region labeled with 255; for example, when calculating the loss function, we ignore the uncertain region to avoid negative effects on model training. The specific process is given as follows:

$$P^{(x,y)} = \begin{cases} 0, & \text{if } \max\left(M_c^{x,y}\right) < \tau_1 \\ \mathrm{argmax}\left(M_c^{x,y}\right), & \text{if } \max\left(M_c^{x,y}\right) \geq \tau_2 \\ 255, & \text{otherwise} \end{cases} \tag{2}$$

where C represents the total class in the dataset. In the classification network, the classification results of the middle layer and the final layer are calculated, and the classification loss is calculated through the cross-entropy loss function [55]. In the segmentation network, the pseudolabel is used to guide the segmentation head to segment, and the segmentation probability result is calculated by cross-entropy to construct the segmentation loss L_{seg}. In addition, the framework further improves the performance of the model by incorporating Adalink and SGBR, which will be described in detail in Sections III-B and III-C.

B. Adalink

The oversmoothing issue in ViT will affect the quality of the pseudolabel generated by the model [56]. Toco uses the middle

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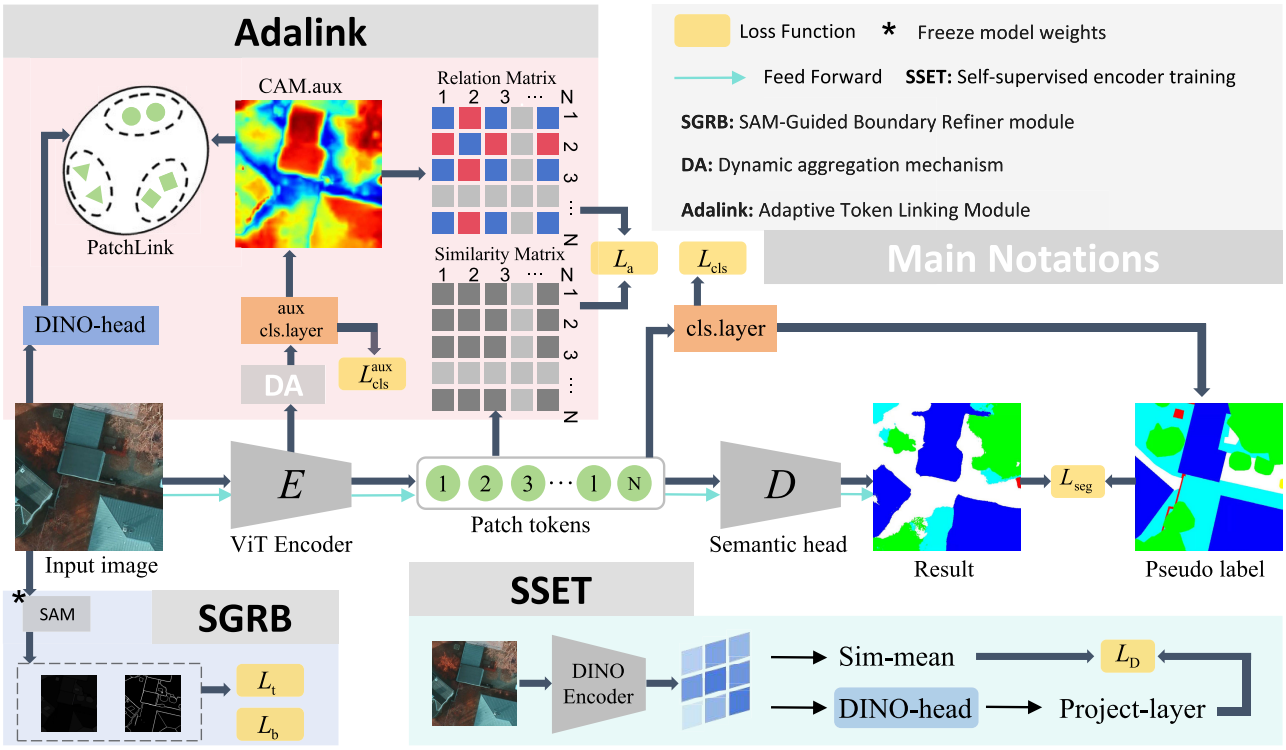


Fig. 2. Overview of our proposed WSSS method, ATSG. In the Adalink model, adaptive extraction of features of the middle layer is combined with self-supervised extraction of token topology relationships. In the SGBR module, a dual-threshold and confidence filtering mechanism is used to suppress the problem of oversegmentation in SAM, thereby extracting more accurate prior information about the target and boundaries.

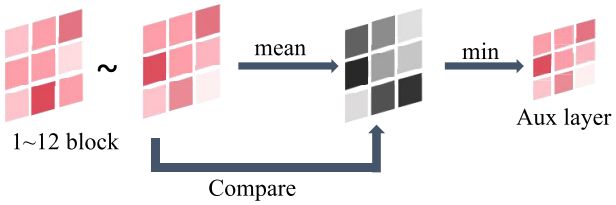


Fig. 3. Our proposed dynamic aggregation mechanism for adaptive extraction of the middle layer.

layer to alleviate this problem. However, Toco's approach is to directly select a fixed middle layer, which will lead to the problem of a too smooth deep layer or a too local shallow layer caused by the fixed layer in different training periods. At the same time, it only depends on the attention of ViT itself and is vulnerable to noise interference in dense remote sensing scenes. Second, the attention of ViT is easily dominated by large targets, and the token correlation of small targets is weak, which is one of the reasons why the natural image model cannot adapt to remote sensing images better. In this regard, we adopt a dynamic aggregation mechanism to adaptively select the middle layer; the process is shown in Fig. 3 and, at the same time, introduces a self-supervised encoder to provide a robust token topology prior for constraining the spatial consistency of ViT features. Specifically, extract each layer in the ViT to calculate the affinity graph $A_i \in R^L$, where L is the number of ViT layers. The ViT divides the image into $N * N$ patches, and each patch becomes a token through linear projection, which is recorded as $X \in \mathbb{R}^{N^2 \times D}$, where D is the feature dimension. The affinity map of each layer is calculated as follows:

$$A = XX^T. \tag{3}$$

The affinity maps of all layers are aggregated and averaged to get the average affinity map $\overline{A}$

$$\overline{A} = \frac{\sum A_i}{L} \tag{4}$$

and then the difference E_i between each layer and the average value is calculated. The layer with the lowest travel value is dynamically selected as the middle layer with high confidence. The formula is given as follows:

$$E_i = |A_i - \overline{A}| \tag{5}$$

where the affinity map A_i satisfies semipositive definiteness and the average map $\overline{A}$ represents the balanced correlation distribution of each layer. The essence of calculating the difference E_i is to measure the correlation consistency between a single layer and the global layer set, and its value is always nonnegative and limited by the $L2$ norm of X_i (because its maximum eigenvalue of A_i is less than $\|X\|_2^2$). Under the weight decay constraint of the AdamW optimizer, the norm of X_i gradually converges with training iterations, leading to a gradual reduction in the fluctuation range of E_i and eventually stabilizing, satisfying the mathematical definition of convergence.

This adaptive middle layer selection operation is more critical for shallow high-frequency details in small target

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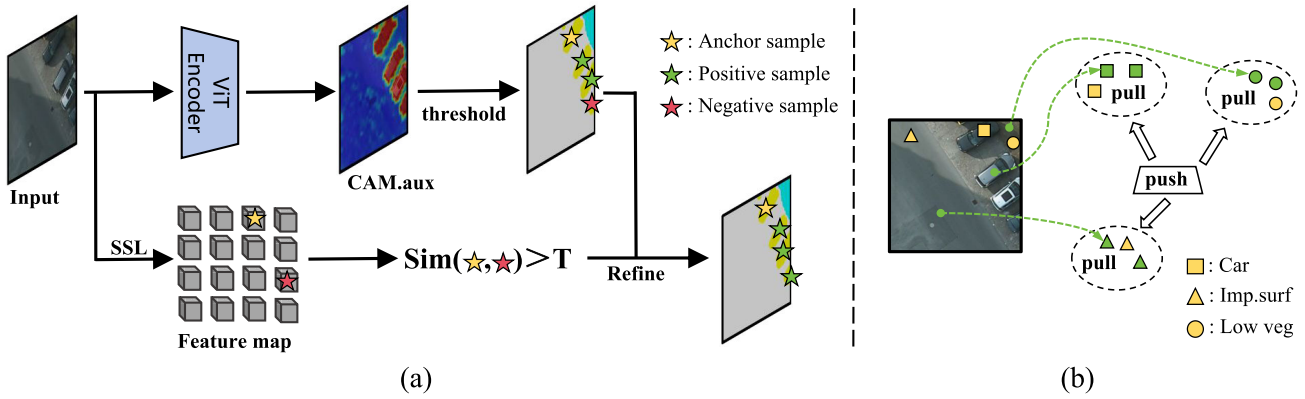


Fig. 4. (a) Process of selecting positive and negative sample pairs. (b) Function of L_a is to make samples of the same class closer in the feature space and push samples of different classes further.

dense scenes, and the layer with the smallest difference may be biased toward the shallow layer. For big goal-dominated scenarios, deep global semantics are more important, and the layer with the smallest difference may be deeper. After the middle layer is extracted, we use the self-supervised encoder to further the topological token relationship. We use the HP [57] method to train a self-supervised encoder in an unsupervised manner. The middle layer patch inputs the self-supervised encoder to obtain the feature F_i^{ssl}. Then, according to the feature F_i^{ssl} extracted by the self-supervised encoder and the features of the middle layer itself, we select positive and negative pairs to construct the L_a loss

$$\mathcal{L}_a = \frac{1}{N^+} \sum_{Y_i = Y_j} \left(1 - \text{CosSim}\left(F_i, F_j\right)\right) + \frac{1}{N^-} \sum_{Y_i \neq Y_j} \text{CosSim}\left(F_i, F_j\right) \tag{6}$$

where N^+ and N^-, respectively, represent the number of positive and negative pairs, F_i is the patch feature output by the ViT encoder from the middle layer, and the construction of positive and negative pairs is first determined by the self-supervised encoder. The process of selecting positive and negative sample pairs is shown in Fig. 4. First, determine whether the similarity between F_i^{ssl} and F_j^{ssl} in the feature space of the self-supervised encoder is greater than the batch average similarity T, and if it is greater, it is considered positive pairs; If it is less than, further judgment is made based on whether the semantic labels P_i^m and P_j^m of the middle layer are the same. If they are the same, it is still considered a positive pair; otherwise, they are negative pairs as follows:

$$\begin{cases} Y_i = Y_j, & \text{if } \left(\text{Sim}\left(F_i^{\text{ssl}}, F_j^{\text{ssl}}\right) > T\right) \cup \left(P_i^m = P_j^m\right) \\ Y_i \neq Y_j, & \text{otherwise} \end{cases} \tag{7}$$

where P^m represents the semantic label converted by the middle layer CAM. The batch average T is publicized as follows:

$$T = \frac{1}{|P|} \sum_{(k,l) \in P} \text{Sim}\left(F_k^{\text{ssl}}, F_l^{\text{ssl}}\right) \tag{8}$$

where $P = \{(k,l) | 1 \leq k \leq l \leq hw\}$ represents the set of all different patch pairs in a batch.

C. SAM-Guided Boundary Refine

In recent years, the development of large models has been particularly rapid. SAM is a general semantic segmentation model, which supports one-click segmentation and other interactive methods. This article introduces SAM to obtain segmentation prior information to obtain high-quality segmentation results. We introduce the target consistency loss function to improve the quality of the pseudolabel by strengthening the semantic consistency of features in the same target area. Specifically, for the target region, this article first calculates its internal central feature vector as the semantic representation of the region. It can be expressed as follows:

$$F_{\text{avg}}^i = \frac{1}{N^i + 1} \sum_{j \in M^i} F^j \tag{9}$$

where F_{avg}^i is the feature vector of the i target area, M^i is the set of all pixels of the i target area in the pseudolabel, and N^i is the number of all feature vectors in the area. In order to mitigate the fluctuation caused by the small number of N^i, we add one to the denominator to ensure the stability of the calculation. In order to avoid the generated segmentation result being too complex [58], this article optimizes the region division by setting a dual-threshold constraint: one is the threshold of the number of segmentation regions T_1 to control the semantic complexity, and the other is the minimum area threshold T_2 to filter out unstable small regions. In the preprocessing stage, the area that meets the threshold value is regarded as an independent target, and a target prior is established. The unassigned area is marked as 0, and the effective area is assigned by number. At the same time, the edge is extracted to construct a boundary prior, where the boundary pixel value is 255 and the nonboundary value is 0. This mechanism not only retains the semantic structure information of salient areas but also enhances the ability of boundary recognition through clear contour features. At the same time, we introduce the CAM as the basis for confidence evaluation to screen out low-reliability areas. Specifically, first, obtain the average value of each type of activation value of the middle layer type activation map for the target area i as the credibility evaluation value α_i^c, which

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is expressed as follows:

$$\alpha_i^c = \frac{1}{|M^i|} \sum_{(x,y)\in M^i} F_{x,y}^c \tag{10}$$

where $|M^i|$ represents the number of pixels in the ith target area and $F_{x,y}^c$ represents the active value of class c at coordinates (x, y). Thus, a mask can be obtained according to the confidence evaluation value, and the target area with a low confidence value can be ignored to avoid the interference of noise on the training process, which is expressed as follows:

$$\hat{S}_{x,y} = \begin{cases} 0, & \text{if } \max \alpha_i^c < \tau \\ 1, & \text{otherwise} \end{cases} \tag{11}$$

where x and y represent pixel coordinates, τ represents the ignore threshold, and $\hat{S}$ represents the generated mask. If the average activation value of all classes in an area is less than the threshold value, it is considered that the area is not reliable enough, and its mask is designed as 0 to avoid introducing this part to disrupt the learning process during subsequent calculation losses, thus effectively reducing noise interference, improving the credibility of the semantic tag acquisition feature relationship, and ensuring the quality of the final pseudolabel. After that, the target consistent loss function can be calculated, which is expressed as follows:

$$L_t = \sum_{i=1}^{K} \left(F_{\text{avg}}^i, F^i\right) \otimes \hat{S} \tag{12}$$

where mse represents the mean square error function and F^i represents all feature vectors in the ith target area. The model can effectively optimize the feature space distribution through the constraint of target consistency loss. On the other hand, Bokhovkin and Burnaev [59] pointed out that using boundary constraints can improve the performance of remote sensing image semantic segmentation. Finally, reconstruction indicators can be obtained, and the boundary loss is defined as follows:

$$\text{BF}_1^b = \frac{2P^bR^b}{P^b + R^b} \tag{13}$$

$$L_b = 1 - \text{BF}_1^b \tag{14}$$

where P^b and R^b are proposed by Bokhovkin and Burnaev [59], which, respectively, represent the boundary accuracy and recall rate. Through the target boundary loss, the model can generate a more compact feature representation and enhance the overall quality of segmentation results with the help of prior knowledge constraints.

IV. Experiments

We conducted extensive experiments on the ISPRS Potsdam and Vaihingen datasets [60] and the iSAID dataset [61] in the field of classical remote sensing semantic segmentation. Compare it with other methods. The experimental results confirm that our method can reach the current SOTA. Simultaneously, conduct ablation experiments to demonstrate the effectiveness of the modules involved.

A. Datasets

1) ISPRS Potsdam Dataset: The ISPRS Potsdam dataset contains 38 high-resolution aerial images of Potsdam, Germany, with a spatial resolution of 5 cm and a size of 6000×6000 pixels per image. It covers six types of land features: impermeable surfaces, buildings, low vegetation, trees, vehicles, and clutter. According to the official classification criteria, 23 were used for training, 14 for validation, and seven to ten were removed due to labeling issues. For ease of training, we cropped the original image into nonoverlapping subgraphs of 256×256 pixels, resulting in 13 248 training samples and 8064 validation samples.

2) ISPRS Vaihingen Dataset: The Vaihingen dataset contains 33 high-resolution aerial images of the German town of Vaihingen, with a spatial resolution of 9 cm and a size of 2494×2064 pixels per image. It covers six types of land features: impermeable surfaces, buildings, low vegetation, trees, vehicles, and clutter. According to the official data classification standard, 16 images are used for training, and 17 images are used for validation. For ease of training, similar to the Potsdam region, we cropped the images into 256×256 nonoverlapping images, resulting in 1986 training images and 752 validation images.

3) iSAID Dataset: The iSAID dataset is a large-scale and annotation intensive dataset for aerial imagery, consisting of 2806 high-resolution images, 655 451 object instances, and a total of 16 classes, including BG, plane (PL), baseball diamond (BD), ship (SH), ground track field (GTF), basketball court (BC), soccer ball field (SBF), roundabout (RA), storage tank (ST), bridge (BR), large vehicle (LV), tennis court (TC), helicopter (HC), harbor (HA), small vehicle (SV), and swimming pool (SP). Similarly, we cropped a 512×512 image from each image. Finally, 7500 training images, 1653 validation images, and 1315 test images were obtained. This dataset supports tasks such as semantic segmentation, instance segmentation, and object detection, offering challenges such as class imbalance, complex BGs, and fine-grained details. It is widely used for advancing aerial image analysis in remote sensing and computer vision.

B. Evaluation Metrics

Based on the above dataset, we use the mean intersection union (mIoU) as an evaluation metric to assess the experimental results as follows:

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{n} \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{15}$$

where TP, FP, and FN represent the number of true positives, false positives, and false negatives, respectively.

C. Implementation Details

For the self-supervised encoder, we utilize the DINO-pretrained ViT-base model [45], which is further trained on remote sensing images to develop a segmentation head for token topology learning in Adalink. The middle layer is dynamically selected through our aggregation mechanism,

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TABLE I
COMPARISON OF SEGMENTATION RESULT ON THE POTSDAM DATASET WITH OTHER METHODS

Method	Imp.Surface	Building	Low veg	Tree	Car	Clutter	mIoU	Training Time
Toco [28]	53.45	50.89	47.72	45.21	48.21	6.35	41.97	**147 mins**
Dupl [38]	53.67	65.32	45.17	**60.97**	46.82	6.97	46.49	480 mins
CTFA [29]	61.55	54.26	55.76	47.81	52.42	11.04	47.14	158 mins
WeCLIP+ [39]	61.22	55.46	54.38	41.96	51.87	6.78	45.28	163 mins
MoRe [30]	63.50	63.18	49.47	39.45	58.62	8.94	47.19	189 mins
Ours	**65.15**	**72.49**	**56.01**	56.78	**59.86**	**12.01**	**53.72**	452 mins

TABLE II
COMPARISON OF PSEUDOLABEL ON THE POTSDAM AND VAIHINGEN DATASETS WITH OTHER METHODS

Dataset	Method	Imp.Surface	Building	Low veg	Tree	Car	Clutter	mIoU
Potsdam	Dupl [38]	50.89	67.41	45.31	**60.95**	54.42	8.82	47.97
	CTFA [29]	59.48	52.74	56.03	55.39	49.43	14.45	47.92
	WeCLIP+ [39]	60.41	53.89	51.80	40.57	50.30	7.98	44.16
	MoRe [30]	**64.62**	64.66	55.40	41.08	56.14	11.62	48.92
	Ours	62.62	**68.69**	**56.68**	59.24	**56.66**	**15.73**	**53.27**
Vaihingen	Dupl [38]	47.09	66.39	26.44	51.02	30.72	1.20	37.14
	CTFA [29]	55.41	61.82	38.24	**58.51**	**40.24**	3.33	42.93
	WeCLIP+ [39]	57.78	61.44	37.32	42.98	25.03	1.87	37.74
	MoRe [30]	57.74	69.13	37.62	42.23	26.14	2.12	39.16
	Ours	**60.06**	**72.51**	**49.46**	53.00	32.51	**3.77**	**45.22**

with detailed analysis of layer selection across different training stages provided in Section IV-E. The SGBR module employs Meta's SAM[1] to generate high-quality object boundary priors. We carefully balance segmentation granularity-excessively fine priors fragment natural objects into numerous small targets, while overly coarse priors merge distinct objects. The optimal segmentation threshold is determined through sensitivity analysis as demonstrated in Section IV-E. Our implementation uses ViT-B as a baseline, trained with AdamW optimizer for 15 000 iterations on Potsdam and Vaihingen datasets and 30 000 iterations on iSAID. The configuration includes a weight decay of $1e^{-2}$, a batch size of 8, a momentum of 0.9, and an initial learning rate of $4e^{-5}$ with polynomial decay. During inference, multiscale CAMs are generated using parameter ratios (1.0, 0.5, 1.5) and fused through averaging, with final refinement via CRF [48]. All experiments were conducted on two NVIDIA RTX 4090 GPUs running Ubuntu 18.04 with PyTorch 2.1.1 and Python 3.8.

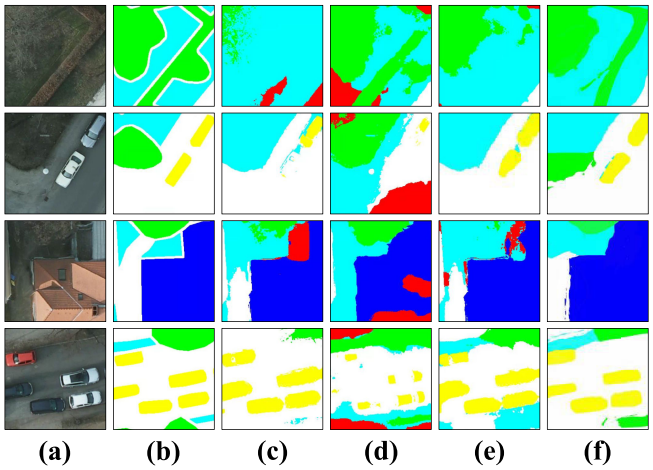


Fig. 5. Visualization results of the WSSS methods on the Potsdam datasets. (a) Input image. (b) Ground truth. (c) Toco. (d) Dupl. (e) MoRe. (f) Ours.

D. Comparison With SOTA Methods

In this section, we compare with other WSSS methods, including MoRe, CTFA, Dupl, and Toco, which are recent SOTA methods. In the experiment, we adjusted the learning rate and other hyperparameters applicable to the remote sensing image dataset to achieve optimal performance of the model. We evaluated the segmentation results of the model and the quality of the generated pseudolabel to validate the effectiveness of our method.

1) ISPRS Potsdam Dataset: The segmentation performance on the Potsdam dataset is compared with other methods in Table I. Our model achieves a 6.53% improvement in mIoU over MoRe, with most classes showing enhanced performance. Notably, the building class demonstrates particularly significant gains due to SAM's strong sensitivity to boundary information, which precisely captures the well-defined edges of structures. The pseudolabel quality comparison in Table II shows that our method outperforms MoRe by 4.35% in mIoU. The visualization results are shown in Fig. 5. Compared with other methods, both the first and third rows show false activation of the cluster class. Our model benefits from the token topology of the Adalink module, which effectively suppresses the noise in the generation of pseudolabel, resulting

[1]SAM:https://github.com/facebookresearch/segment-anything

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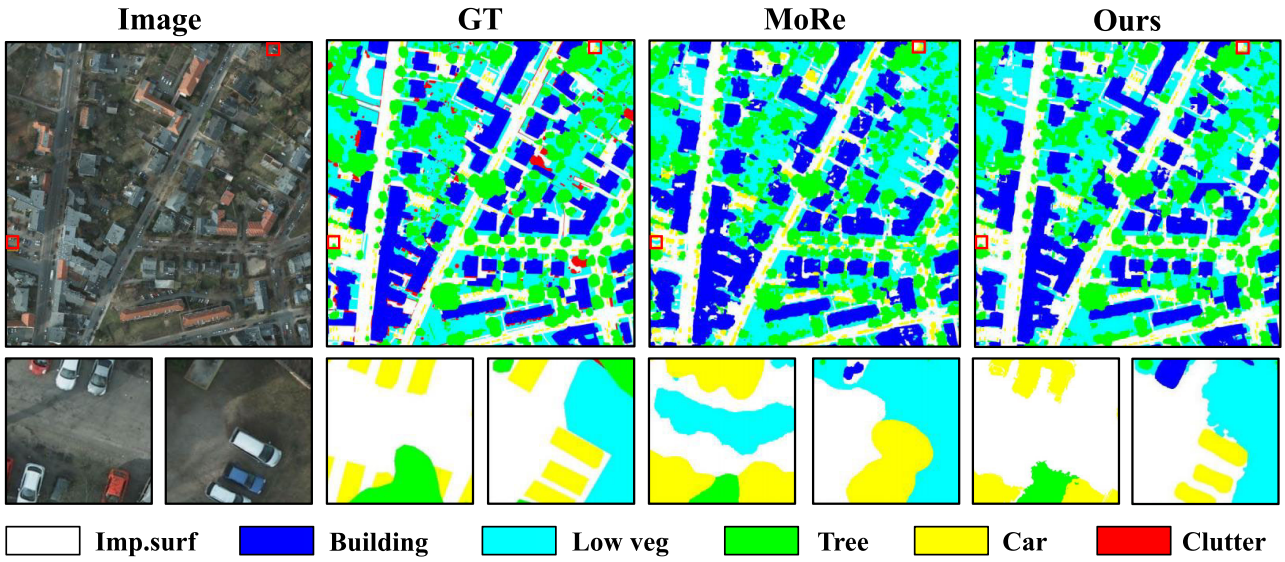


Fig. 6. Examples of segmentation results from WSSS methods on the ISPRS Potsdam dataset.

TABLE III
COMPARISON OF SEGMENTATION RESULT ON THE VAIHINGEN DATASET WITH OTHER METHODS

Method	Imp.Surface	Building	Low veg	Tree	Car	Clutter	mIoU	Training Time
Toco [28]	52.07	43.77	29.24	52.41	24.57	1.63	33.95	**41 mins**
Dupl [38]	52.31	68.89	32.74	53.12	20.29	1.23	38.10	142 mins
CTFA [29]	59.13	64.02	38.16	**58.06**	32.26	2.35	42.33	51 mins
WeCLIP+ [39]	56.69	63.02	36.97	46.96	27.88	1.64	38.86	49 mins
MoRe [30]	58.84	68.14	38.41	52.79	21.17	2.15	40.25	59 mins
Ours	**60.44**	**73.01**	**50.05**	52.24	**34.71**	**6.91**	**46.23**	96 mins

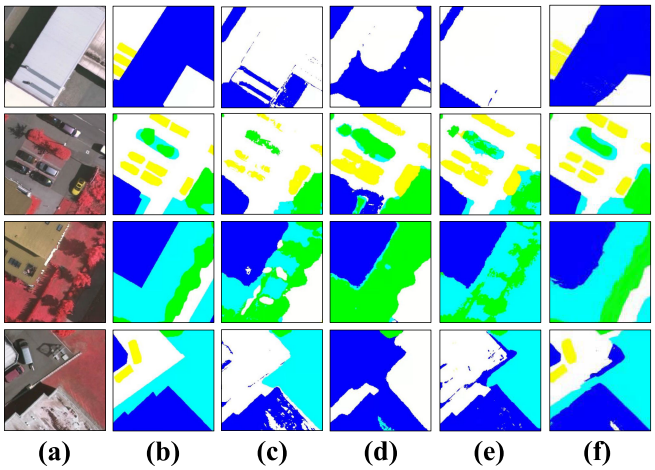


Fig. 7. Visualization results of the WSSS methods on the Vaihingen datasets. (a) Input image. (b) Ground truth. (c) Toco. (d) Dupl. (e) MoRe. (f) Ours.

in more accurate segmentation results. In addition, in the second row, tree and low vegetation appear at the same time, causing other methods to mistakenly integrate them or merge them with the surrounding classes, resulting in segmentation errors. Our method successfully divides these two classes, showing good robustness. For comprehensive evaluation, we reconstructed local predictions into full-scale images matching the original data dimensions, as shown in Fig. 6. Comparative analysis reveals that MoRe suffers from false segmentation of low vegetation and other errors. Specifically, in vehicle segmentation, MoRe incorrectly merges the interstitial spaces between adjacent vehicles into single instances. In contrast, our SAM-enhanced approach demonstrates superior boundary recognition, accurately distinguishing neighboring vehicles while maintaining complete, individual segmentation results.

2) ISPRS Vaihingen Dataset: The segmentation performance on the Vaihingen dataset is compared with other methods in Table III. Our method achieves a 3.9% improvement in mIoU over CTFA, with enhanced performance observed in most classes. The pseudolabel quality comparison in Table II demonstrates a 2.29% mIoU gain compared to CTFA. The visualization results are shown in Fig. 7. Compared with other methods, in the first row, a car is located on the shaded side of a building due to obstructed lighting, making its appearance very similar to the surrounding shadows. While other methods misclassify this region (including the car) as an impervious surface, our model accurately segments the car, owing to SAM's capability to precisely localize small objects. In the third example, a tree casts a shadow under strong illumination. Existing methods incorrectly segment both the tree and its shadow as a single tree class, even missegmenting parts

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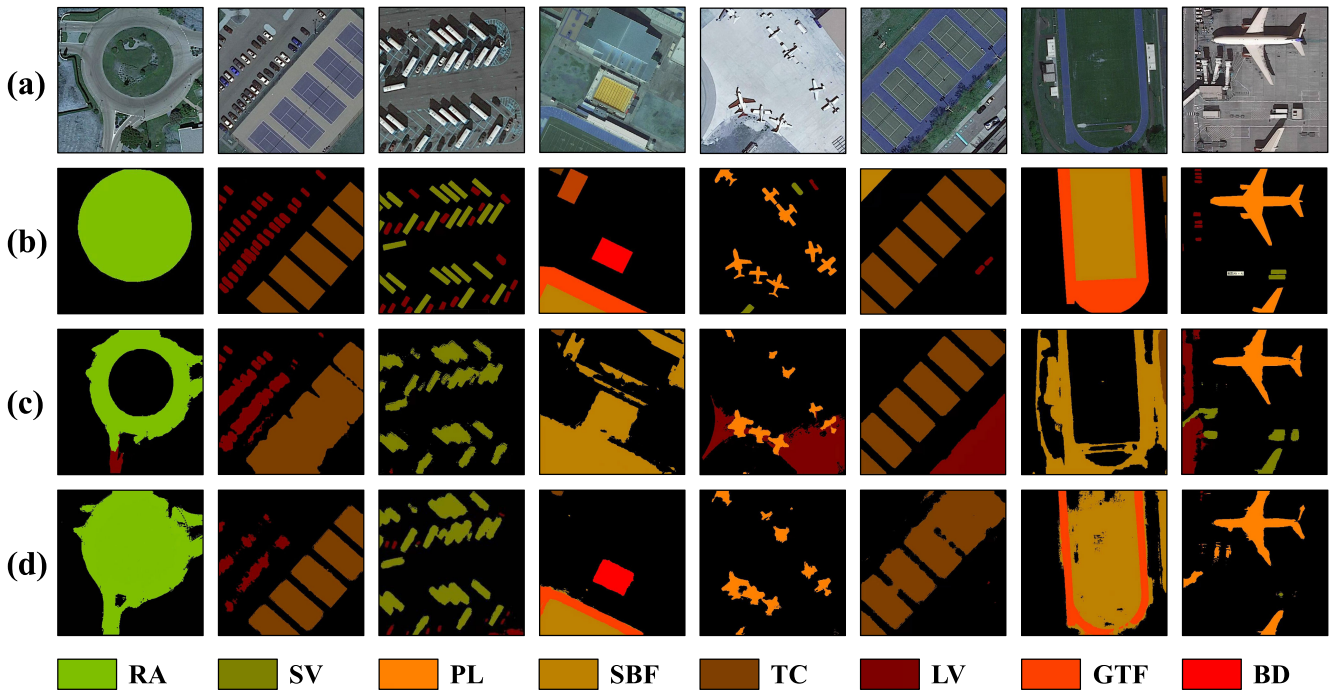


Fig. 8. Visualization results of the WSSS methods on the iSAID datasets. (a) Input image. (b) Ground truth. (c) MoRe. (d) Ours.

TABLE IV
COMPARISON OF SEGMENTATION RESULT ON THE iSAID DATASET WITH OTHER METHODS

Method	BG	GTF	SBF	SV	SH	BR	BC	BD	RA	PL	TC	LV	ST	HA	SP	HC	mIoU
CLIMS [62]	82.41	2.49	11.39	5.21	3.78	8.41	16.82	14.56	18.30	27.33	7.68	19.49	4.47	11.25	26.24	3.47	16.46
Mctformer [63]	76.63	10.40	35.67	2.79	16.10	7.51	39.83	39.98	15.66	**55.31**	35.70	17.11	41.10	10.31	34.45	3.31	27.62
CISM [64]	83.10	33.50	44.60	11.80	19.50	8.40	48.30	34.40	29.20	30.50	52.60	31.90	44.10	10.80	17.60	14.30	32.16
Toco [28]	90.17	54.60	39.52	11.85	27.98	14.93	29.55	38.94	13.56	37.44	48.11	33.20	39.46	32.46	32.22	23.24	35.45
OME [43]	**94.62**	36.60	**52.72**	13.12	22.53	13.12	32.49	9.87	**31.15**	48.38	**66.22**	**54.73**	43.03	31.31	8.17	16.99	35.94
Dupl [38]	81.36	55.73	21.82	8.22	19.83	2.70	34.85	30.12	8.31	39.08	64.41	33.60	34.54	10.16	37.28	11.33	30.83
CTFA [29]	90.20	55.97	40.18	14.87	28.73	14.97	42.12	38.21	20.15	37.72	50.67	35.72	48.15	32.85	32.82	22.10	37.84
WeCLIP+ [39]	88.14	22.80	31.74	7.34	10.12	10.74	28.96	36.13	25.64	39.57	29.78	15.31	30.87	10.29	11.62	5.98	25.31
MoRe [30]	87.34	60.90	39.03	9.70	31.93	**23.73**	42.09	54.07	17.66	39.37	46.44	33.89	**55.52**	18.86	24.81	**25.15**	38.16
ISANet [65]	91.10	18.70	24.90	14.30	13.50	1.00	**49.70**	**66.40**	13.70	51.00	62.30	40.30	51.00	27.50	40.70	34.20	37.52
Ours	92.35	**65.92**	48.67	**15.06**	**32.46**	11.15	29.73	46.18	9.96	41.22	62.71	36.49	43.61	**35.67**	**47.56**	12.19	**39.43**

TABLE V
COMPUTATIONAL COST COMPARISON ON THE iSAID DATASET

Method	All Params	Training Time	Inference Time
Toco [28]	105M	**164 mins**	**74 s**
Dupl [38]	185M	530 mins	102 s
CTFA [29]	112M	179 mins	95 s
WeCLIP+ [39]	456M	189 mins	117 s
MoRe [30]	**95M**	307 mins	98 s
Ours	277M	359 mins	88 s

of the shadowed area. In contrast, our method successfully distinguishes and correctly segments this region.

3) isaid Dataset: The segmentation results on the iSAID dataset compared to other methods are shown in Table IV. Our method improved mIoU by 1.27% compared to MoRe, and the visualization results are shown in Fig. 8. In the first column, the MoRe method mistakenly identified the middle area of the RA as the BG. In the second column, the MoRe mistakenly connected several adjacent TCs, and our method benefited from the SGBR module, dividing several adjacent TCs into separate targets. In the third column, MoRe hardly segmented SVs. The target of this class is very small, so it is difficult to segment. Our method successfully segmented some SVs. In the seventh column, our method effectively solved the problem of similar foreign matter with the same spectrum, successfully distinguished the runway, the stadium, and the off-site area, and avoided the error of mistaking the outer ring as the stadium and the infield as the BG. In the last column of Fig. 8, more mistakenly segmented the aircraft wing below into LVs, but our method maintains good robustness. Our method has a large number of parameters, as shown in Table V, reflecting the increased structural complexity

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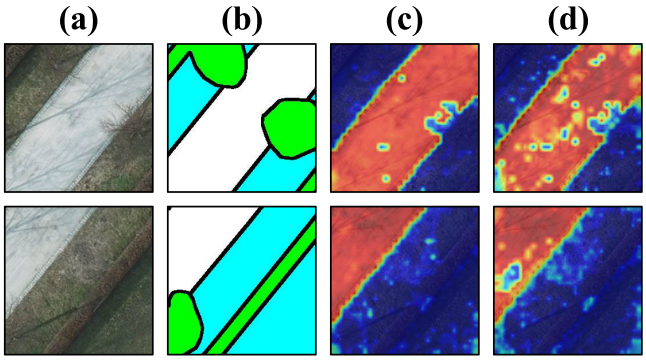


Fig. 9. Comparison of CAM in the middle and last layers. (a) Input image. (b) Ground truth. (c) Middle CAM. (d) Final CAM.

TABLE VI
ABLATION STUDY OF THE PROPOSED MODULES ON THE POTSDAM DATASET

Baseline	SGBR	Adalink	mIoU	Aux mIoU	Car mIoU
✓	None	None	39.28	33.16	47.33
✓	None	✓	49.51	40.98	53.85
✓	✓	None	47.54	36.07	57.12
✓	✓	✓	**53.72**	**42.54**	**59.86**

introduced by SAM. Despite this scale, our method achieves a training time significantly lower than that of Dupl, another large-scale model, demonstrating training efficiency under high parameter counts. Furthermore, in terms of inference speed, our model performs competitively, achieving inference times on par with several leading approaches—including some with notably fewer parameters—highlighting its practical suitability for deployment scenarios where both capacity and latency are considered.

E. Ablation Studies

In order to further verify the effectiveness of our method, we conducted ablation experiments on the remote sensing dataset Potsdam and conducted ablation research on the main modules and some super parameters we proposed, in order to explore how these modules affect the segmentation results and reasonably set the super parameters.

1) Token Topology: The ablation study validates the effectiveness of our Adalink module in modeling token relationships within the middle layer. As demonstrated in Table VI, our proposed Adalink module achieves a 10.23% mIoU improvement over the baseline method on the Potsdam dataset. The key advantage lies in the encoder's ability to precisely identify semantic boundary regions by capturing spatial topological relationships between tokens, including adjacency continuity and boundary transition characteristics. Fig. 9 shows the CAM of the impervious surface class during the mid-training phase, where the middle layer produces more complete and accurate object activation compared to the final layer. The Adalink demonstrates particular effectiveness in discriminating between genuine semantic boundaries and mere feature response fluctuations. Our experiment ultimately demonstrated that introducing the Adalink to topology token relationships not only retains the advantage of baselines in utilizing middle features but also significantly improves the geometric accuracy and semantic coherence of pseudolabels.

2) Dynamic Aggregation Mechanism: Our systematic experimental analysis reveals significant insights into the relationship between middle layer selection and training stages through the dynamic aggregation mechanism. Fig. 10 demonstrates that at the initial training stage (iter 3000), the model preferentially selects the shallow middle layer (L3), which preserves rich spatial details and low-level features [28], providing stable foundations for pseudolabel generation. Experimental data show that the L3 layer achieves 36.12% at iter 3000, substantially outperforming Toco's fixed L9 layer selection (24.11%). As training progresses to iter 6000, the model transitions to a medium-depth layer, which balances spatial information with emerging semantic features, forming an optimal middle state. Notably, during mature training stages (post iter 12 000), the deeper layer demonstrates clear advantages: L7 achieves peak performance (47.48%) at iter 12 000, while L9 becomes optimal (51.38%) at iter 15 000. This "shallow-to-deep" progression mirrors human visual cognition, empirically validating the concrete-to-abstract feature learning paradigm in deep models. Our method achieves a mIoU of 53.72%, outperforming the fixed-middle layer baseline (51.38% at 15k iter) by 2.34%. This gain highlights its adaptive superiority. This approach overcomes the limitation of using fixed middle layers by adaptively selecting more suitable layers during different training phases. In the early training stage, it avoids interference from deep feature noise; during the middle phase, it achieves a smooth transition, while, in the later stage, it can fully utilize deep semantic information.

3) SAM-Guided Boundary Refine: Our ablation study thoroughly examines the impact of SAM-generated boundary maps and target maps on final segmentation performance. As evidenced in Table VI, the SGBR module yields an 8.26% mIoU improvement on the Potsdam dataset. The boundary maps excel at enhancing geometric precision along segmentation edges, particularly for structured objects, while the target maps substantially improve intraclass consistency through global object priors. Notably, removing SAM supervision most severely impacts small object segmentation, with car class performance dropping by 9.79%, confirming SAM's crucial role in preserving fine-grained details. The synergistic combination of our two main modules achieves a 14.44% quality improvement over baseline. These results conclusively demonstrate that SAM's multilevel visual cues and ViT features form complementary advantages, collectively advancing weakly supervised segmentation performance. We also visualized the impact of two modules. In Fig. 11, without adding two modules, the building only activated its most discriminative area, while there were some erroneous activations of cars and trees along with the BG. In the building class, after adding the SGBR module, due to the constraint ability of SAM, the building was fully activated, and there was a strong constraint ability at the boundary. Only adding the Adalink module, the activation area was more complete while significantly suppressing BG noise with low association with the building. After the addition of two modules simultaneously,

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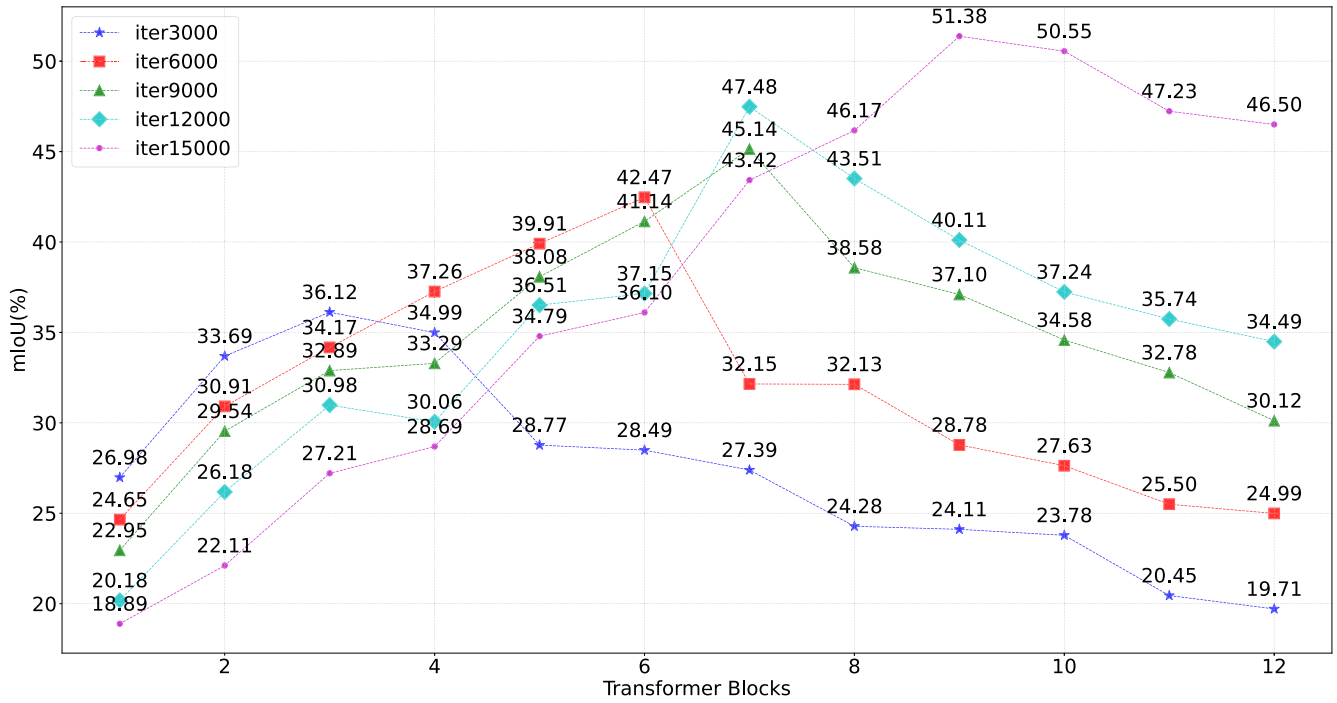


Fig. 10. mIoU of 12 transformer blocks under different iterations. The model performance improves progressively with deeper layers as training iterations advance.

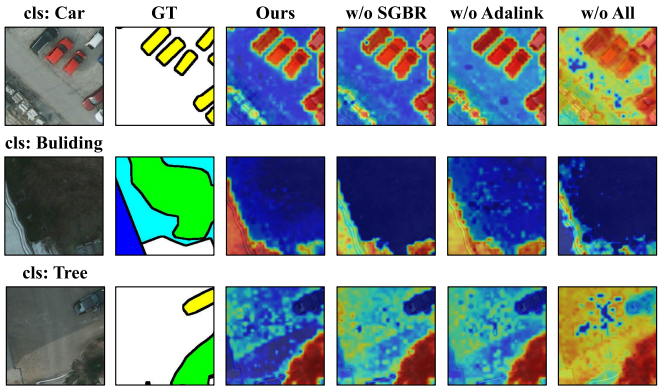


Fig. 11. Visual CAM of each module. Each row displays a specific class, and the last three columns show the removal of the SGBR module, the Adalink module, SGBR, and the Adalink module.

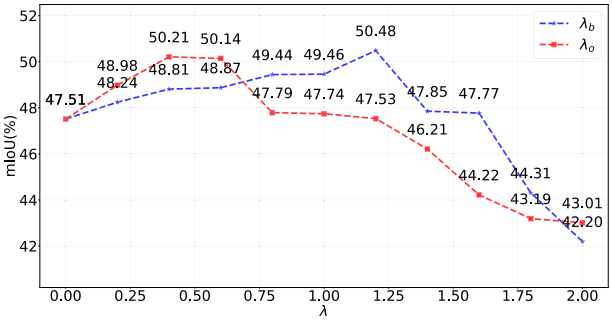


Fig. 12. Sensitivity analysis of λ_b and λ_t.

TABLE VII
SENSITIVITY ANALYSIS OF T_1 AND T_2 ON THE POTSDAM DATASET

$T_1 \setminus T_2$	50	100	150	200
15	49.24	49.85	48.92	48.51
30	49.81	50.48	50.42	50.38
45	50.03	**53.72**	51.31	51.10
60	49.83	50.60	50.39	50.33

the activation of the building is complete, the target contour is accurate, and the BG noise is suppressed. This is due to Adalink's enhanced activation signal and BG noise suppression functions, as well as the core value of SGBR, which is to optimize class boundaries and complete target contour functions, resulting in strong model performance.

4) Hyperparameter:

a) Boundary and target loss weights: As illustrated in Fig. 12, experimental results demonstrate optimal model performance when $\lambda_b = 1.2$. Excessive λ_b values lead to overreliance on potentially inaccurate boundary priors, particularly problematic in scenes with ambiguous boundaries such as transitional areas between grass and ground or building shadows caused by lighting. In cases where SAM-generated boundaries contain fractures or offsets, overly strong supervision propagates these errors to the final segmentation. Conversely, insufficient λ_b values prevent effective utilization of accurate boundary information. For λ_t, the optimal value is 0.4. Overly large λ_t values cause the model to overfit the homogeneity assumption within target regions, adversely affecting objects with internal structures or uneven intensity distributions. Insufficient λ_t values, however, limit the model's ability to leverage valuable object prior information.

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TABLE VIII

SENSITIVITY ANALYSIS OF PSEUDOLABEL CONVERSION THRESHOLD ON DIFFERENT DATASETS. THE ORDER IS THE POTSDAM DATASET, THE VAIHINGEN DATASET, AND THE ISAID DATASET

		β_l			
		0.15	0.2	0.25	0.3
β_h	0.6	51.3	51.4	52.8	50.3
	0.65	51.4	51.6	53.1	52.3
	0.7	52.9	53.4	**53.7**	52.6
	0.75	52.1	52.3	53.0	50.1

		β_l			
		0.15	0.2	0.25	0.3
β_h	0.6	44.1	44.6	45.3	44.3
	0.65	44.3	45.3	46.1	44.8
	0.7	45.1	46.0	**46.2**	45.4
	0.75	44.2	44.5	44.9	44.0

		β_l			
		0.15	0.2	0.25	0.3
β_h	0.6	36.5	37.2	37.7	37.1
	0.65	37.2	38.5	38.7	38.3
	0.7	37.5	39.2	**39.4**	38.7
	0.75	37.2	37.4	38.3	36.2

TABLE IX

COMPARISON OF DIFFERENT LOSS FUNCTIONS FOR TARGET CONSISTENCY LOSS ON THE POTSDAM DATASET

Loss Function	Imp.Surface	Building	Low veg	Tree	Car	Clutter	mIoU
CE Loss	62.02	68.12	54.79	53.81	53.25	9.47	50.24
Dice Loss	62.11	69.94	55.21	54.18	55.23	10.17	51.14
MSE Loss(Ours)	**65.15**	**72.49**	**56.01**	**56.78**	**59.86**	**12.01**	**53.72**

b) Dual threshold in the SGBR module: The dual thresholds in the SGBR module are configured based on dataset statistics, including the average region number and the minimum target size. As demonstrated in Table VII, the optimal performance was obtained with $T_1 = 45$ and $T_2 = 100$. Here, T_1 governs the number of segmented regions by controlling the merging process. A higher value of T_1 results in more aggressive merging, producing fewer regions with high-level, coarser semantics. Conversely, T_2 serves as a minimum area threshold to eliminate unstable small regions. Its variation primarily influences the trade-off between detail preservation and noise suppression—a larger T_2 value filters out more small regions, retaining only larger and more stable ones.

c) Threshold in pseudolabel conversion: In order to investigate the impact of high and low thresholds on model performance during pseudolabel generation, we conducted detailed sensitivity analysis on three remote sensing image segmentation datasets: Potsdam, Vaihingen, and iSAID. As shown in Table VIII, the pseudolabel conversion threshold has a significant impact on model performance and exhibits similar patterns on the three datasets. When β_h is 0.7 and β_l is 0.25, the model achieves optimal performance on all three datasets, confirming the robustness and universality of this parameter combination.

d) Target consistency loss selection: As shown in Table IX, the mIoU of mse loss reached 53.72%, higher than Dice loss by 2.58% and higher than CE loss by 3.48%. This indicates that mse has better constraints on the consistency of target region features. In contrast, although Dice loss has traditional advantages in optimizing overlap at the mask level, its control effect on global feature distribution is weaker than mse in the feature space constrained scenario of this method, resulting in slightly poorer overall performance. Second, mse can make the feature distribution of the same target area more compact, manifested as a higher IoU of the car, which is particularly critical when segmenting small targets. This not only helps to maintain the integrity of the target but also effectively suppresses the interference of BG noise on the target features.

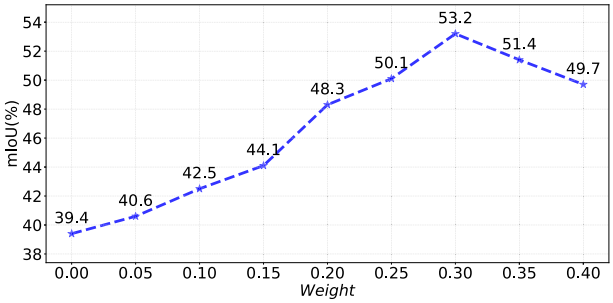


Fig. 13. Sensitivity analysis of the weight of loss L_a.

5) Contrastive Loss L_a: As shown in Fig. 13, the loss function L_a plays an important role in the model. When its weight is set to 0.3, mIoU increases to 53.2%, which is 13.8% higher than the benchmark. Too small weight weakens the supervisory role of the middle layer features on the final output, resulting in the model being unable to fully utilize the contrastive learning signals between hierarchical features. On the other hand, too large weight can disrupt the optimization direction of the main loss function, causing the model to overly focus on the contrastive relationship between feature layers and ignore the final segmentation accuracy. Therefore, setting it around 0.3 will effectively improve the model performance.

6) Standard Deviation Analysis: We conducted standard deviation experiments on the Potsdam and Vaihingen datasets under the same conditions, running ten times to calculate the mean and standard deviation. The results are shown in Table X; our method achieved a higher average mIoU (52.49%) on the Potsdam dataset while demonstrating a stable standard deviation ($\pm$1.12%). The standard deviations of various methods are basically at the same level, and our method is 6.53% higher than MoRe, demonstrating strong performance improvement. The results indicate that our proposed ATSG framework not only improves segmentation accuracy but also, more importantly, has strong model robustness and stability.

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TABLE X
COMPARISON OF SEGMENTATION PERFORMANCE WITH STANDARD DEVIATION ON THE POTSDAM AND VAIHINGEN DATASETS

Dataset	Method	Imp.Surf.	Building	Low Veg	Tree	Car	Clutter	mIoU
Potsdam	Dupl [38]	52.11	64.98	44.12	**57.82**	45.51	6.12	45.11 ± 1.09
	CTFA [29]	61.41	52.11	54.54	45.91	51.01	10.23	45.87 ± 0.97
	WeCLIP+ [39]	59.13	54.68	52.24	41.21	49.36	6.21	43.81 ± 1.26
	MoRe [30]	62.27	61.31	47.64	38.18	58.22	8.05	45.95 ± 1.18
	Ours	**63.95**	**69.87**	**55.75**	55.13	**59.02**	**11.21**	$\mathbf{52.49 \pm 1.12}$
Vaihingen	Dupl [38]	50.21	67.58	31.69	51.70	20.14	1.17	36.92 ± 1.01
	CTFA [29]	57.21	61.98	37.49	56.25	31.21	1.93	41.01 ± 1.22
	WeCLIP+ [39]	55.31	62.19	34.59	45.67	24.96	1.07	37.30 ± 1.45
	MoRe [30]	57.12	66.38	37.27	51.76	20.25	1.72	39.08 ± 1.21
	Ours	**59.21**	**71.23**	**49.31**	51.38	**33.32**	**5.21**	$\mathbf{44.94 \pm 1.24}$

V. DISCUSSION

The proposed ATSG has demonstrated SOTA performance on multiple remote sensing datasets; our success is inseparable from the ability of the ViT architecture and basic models to capture complex spatial relationships. SimPoolFormer [4] uses a ViT-based Simpool improvement method for spectral spatial feature extraction, S2C [53] uses SAM's automatic segmentation for prototype comparison learning. These are excellent examples of related fields.

Although this study achieved SOTA on multiple datasets, it must be noted that the model's generalization ability still faces challenges due to the inherent characteristics of remote sensing image datasets. There are significant differences in spatial resolution, spectral characteristics, terrain scale, and imaging conditions among different remote sensing datasets. For example, although the ISPRS Vaihingen and Potsdam datasets are both urban scenes, their architectural styles and vegetation distributions are different, while the iSAID dataset contains more diverse and usually very-small-scale instance targets, and the DeepGlobe dataset [66] mainly targets agricultural areas. Our model, like most current deep learning methods, essentially learns the empirical distribution of training data. Differences in datasets can lead to significant differences in the setting of key hyperparameters for the model. Due to fundamental differences in spatial resolution, terrain scale, target density, and scene complexity among different remote sensing datasets, the optimal hyperparameter combination obtained through careful tuning on one dataset often cannot be directly transferred to another dataset.

Our model also has some limitations, and it has poor segmentation performance for extremely small and rare targets. This limitation is mainly due to the superposition of two factors. First, the severe class imbalance, inherent in the input data, hinders the model from learning robust and discriminative feature representations for these rare classes. Second, although SAM's zero sample segmentation is powerful, it exhibits instability on irregularly small targets. SAM relies on uniform grid point prompts, and for targets with extremely low pixel proportions and irregular shapes, the generated masks often appear highly fragmented or severely incomplete. In addition, due to the introduction of SAM, our method generates a large number of model parameters and increases training time.

In response to the above limitations, future research can be conducted in the following directions. First, to enhance the model's perception ability of extremely small and rare targets, targeted data augmentation strategies and dedicated loss functions can be explored to force the model to pay more attention to tail classes during training. Second, to solve the problem of unstable segmentation of small targets by SAM in fully automatic mode, a feasible path is to study lightweight prompt engineering, such as using the initial CAM to generate rough bounding boxes or point prompts for the target area, in order to guide SAM to generate more complete and reliable target masks.

VI. CONCLUSION

This article presents a new WSSS method, ATSG. By adaptively selecting the most reliable middle layer features at different training stages, our method achieves dynamic feature fusion and effectively overcomes the limitations of fixed middle layer selection. At the same time, we introduce a self-supervised encoder to construct a token association graph, which significantly alleviates the problems of interclass confusion and intraclass differences, thus greatly improving the quality of pseudolabel. In addition, to address the challenge of small target segmentation in remote sensing images, the zero sample segmentation ability of SAM is used to automatically extract the prior of the target boundary, which further improves the positioning accuracy and segmentation integrity of the small target. Extensive experiments on the widely used remote sensing datasets—ISPRS Potsdam, ISPRS Vaihingen, and iSAID—validate the effectiveness of the proposed method and achieve SOTA results. In future work, we will explore the fusion of multimodal data to enable complementary feature learning, with the goal of alleviating false activation in pseudolabel and further improving the model's perception of complex scenes.

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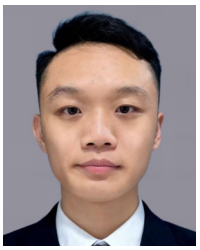
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